AMC 10 Diagnostic Test

Worked solutions · Choose a question to review.

Question 1 · Answer C

Set the two fares equal: 50 + 0.08m = 40 + 0.10m. The $10 difference in starting fees is offset by 2 cents per mile, so m = 10/0.02 = 500 miles.

Question 2 · Answer D

Ten pieces require nine cuts. Cuts 1–5 take place Monday through Friday; cuts 6–9 are Monday through Thursday the following week. The final cut is on Thursday.

Question 3 · Answer B

At 72 minutes Larry has finished nine batches of 3 and Mary six batches of 5, a total of 57. The next finished batch is Larry’s at 80 minutes, bringing the total to 60. Eighty minutes after 3:00 is 4:20 PM.

Question 4 · Answer C

Both triangles use the same altitude to the line BC. D can lie 6 units on either side of C, giving BD = 3 or 15. The corresponding areas are 60 × 3/9 = 20 and 60 × 15/9 = 100. The difference is 80.

Question 5 · Answer B

Let q be the quotient on division by 8. The number equals 8q + 5 and also 5(2q) + 3. Equating gives 2q = 2, so q = 1 and the number is 13.

Question 6 · Answer D

All-odd numbers have five choices in each of four positions: $5^4 = 625$. An all-even four-digit number cannot start with 0, so there are $4 \times 5^3 = 500$. The groups are disjoint, giving 1,125.

Question 7 · Answer D

The quadratic is $(x - n)^2 = 4$, so its roots are $n - 2$ and $n + 2$. The smaller must exceed -3 and the larger must be below 5. Hence $-1 < n < 3$: the integers 0, 1 and 2 sum to 3.

Question 8 · Answer C

Each of n vertices joins to n − 3 nonadjacent vertices, counting each diagonal twice. Thus n(n − 3)/2 = 2015. The positive root is (3 + √16129)/2 = (3 + 127)/2 = 65.

Question 9 · Answer E

There are 12 choose 3 = 220 equally likely triples. Choose one ball of each colour in 3 × 4 × 5 = 60 ways. The probability is 60/220 = 3/11.

Question 10 · Answer B

Factor $(x - y)(x + y) = 31$. The factors are positive integers, so they are 1 and 31. Thus $x = 16$, $y = 15$, and $x^2 + y^2 = 256 + 225 = 481$.

Question 11 · Answer C

Let r be the sphere’s radius and s the cube’s edge. The surface areas give $4\pi r^2 = 12s^2$, so $r/s = \sqrt{(3/\pi)}$. The volume ratio is $(4\pi/3)(r/s)^3 = 4\sqrt{(3\pi)}/\pi$.

Question 12 · Answer A

The class size is one less than a multiple of both 3 and 5. The possibilities are 14, 29, 44, … . The first multiple of 4 is 44.

Question 13 · Answer C

Set D = (0,0), C = (s,0), A = (0,s), B = (s,s) and E = (0,e). CE = 15 + 9 = 24. F is 5/8 of the way from C to E, so F = (3s/8, 5e/8). It lies on y = x, hence e = 3s/5. Continuing the same line to height s multiplies the distance from C by s/e = 5/3. Thus CG = 40 and EG = 40 − 24 = 16.

Question 14 · Answer E

Let T(n) count tilings of a 3-by-n strip. Its first column is either all unit squares, or starts a 2-by-2 square in one of two positions. In the second case the two cells beside the square must be unit squares. Thus $T(n) = T(n-1) + 2T(n-2)$. Starting with $T(0) = T(1) = 1$ gives 3, 5, 11, 21, 43, 85, 171. The required value is 171.

Question 15 · Answer A

Opposite sides of a tangential quadrilateral have equal sums, so each equal leg is (15 + 60)/2 = 37.5. The horizontal projection of each leg is (60 − 15)/2 = 22.5. Its vertical height is $\sqrt{(37.5^2 - 22.5^2)} = 30$. The circle touches both bases, so its diameter is 30 and its radius is 15.

Question 16 · Answer D

The increase in trailing zeroes from $(n-1)!$ to $n!$ equals the number of factors of 5 in n. We need a three-digit multiple of 125 that is not divisible by 625. The largest is $875 = 7 \times 125$. Hence $n-1 = 874$ and its digits sum to 19.

Question 17 · Answer B

For integer side lengths $a,b \geq 1$, $A + P + 4 = (a + 2)(b + 2)$, with both factors at least 3. Adding 4 to the options gives $104 = 8\times13$, $106 = 2\times53$, $108 = 9\times12$, $110 = 10\times11$ and $112 = 8\times14$. Only 106 lacks an allowed factorisation, so $A + P$ cannot equal 102.

Question 18 · Answer A

Factor $n^{2015} - n$ as $n(n^{2014} - 1)$. Its two factors are coprime, so if their product is square and positive, each is a square. For $n > 1$, the second factor is exactly one below $(n^{1007})^2$ and above the preceding square. It cannot be square. The value $n = 1$ works because $0 = 0^2$; there is exactly one.

Question 19 · Answer E

Choose E = (0,0), A = (−3,0), C = (24,0), D = (0,4); ED = 4 follows from AD = 5. Intersecting chords give 3×24 = EB×4, so B = (0,−18). The perpendicular bisectors of AC and BD meet at (21/2,−7). Its distance from E is $\sqrt{((21/2)^2 + 7^2)} = 7\sqrt{13}/2$.

Question 20 · Answer C

Twenty-four elves receive two gifts and sixteen receive one. Choose the 24 elves, then order all gifts and ignore the order inside each pair. $N = C(40,24) \times 64! / 2^{24}$. The exponent of 2 in 64! is 63. In C(40,24) it is 38 − 22 − 15 = 1, by summing multiples of 2, 4, 8, … in the three factorials. The exponent in N is 40; a base-4 zero uses two factors of 2, giving 20.

Question 21 · Answer C

Exterior turns are 60°. Place A = (0,0), B = (1,0), C = (2,√3), D = (1/2,5√3/2). Horizontal and vertical closure, using EF = 2, gives DE = 1 and FA = 3. Thus E = (−1/2,5√3/2) and F = (−3/2,3√3/2). The shoelace cross-products, in units of √3, sum to 0 + 1 + 9/2 + 5/2 + 3 + 0 = 11; half is the area, 11√3/2.

Question 22 · Answer D

The 4,029 replies must be the integers 0 to 4,028. The person who does not know 4,028 people knows only their spouse; the person who knows everyone must be that spouse. Remove this pair: among those left, each unknown-count falls by one. Repeating pairs off the opposite extremes, leaving the man’s wife with the middle original unknown-count, 2,014. She knows 4,029 − 2,014 = 2,015 people.

Question 23 · Answer E

Write the math scores as 85−d, 85, 85+d. The history scores are 83+d, 84, 87−d. The geometric progression gives $(83+d)(87-d) = 84^2$, so $(d-15)(d+11) = 0$. Paul's math score exceeds his history score, forcing $d > 1$ and hence $d = 15$. The math scores run from 70 to 100, a range of 30.

Question 24 · Answer E

Let a(n) = floor(n^2/2015). Through n = 1008, consecutive unrounded terms differ by at most 1 because $(n+1)^2 - n^2 = 2n+1$. The floors therefore take every integer from 0 to floor(1008^2/2015) = 504: 505 values. Each of the remaining 1,007 steps increases the unrounded value by more than 1, so each yields a new floor. Total: 505 + 1,007 = 1,512.

Question 25 · Answer C

Write v = 12k and r = 3k+6. From any one vertex count two-edge walks that finish at a different vertex: there are r(r−1). Each of the other v−1 vertices is reached by exactly c such walks, so c = r(r−1)/(v−1) must be an integer. For the five possible k the ratios are 72/11, 132/23, 210/35, 306/47 and 420/59. Only 210/35 is integral, giving k = 3.